Medical Insurance - ML Report

1. Dataset Overview

Source file : medical_insurance.csv
Rows : 1,338
Columns : age, sex, bmi, children, smoker, region, charges
Avg charges : $17,663.52
Smokers : 269 (20.1%)

2. Descriptive Statistics

index	age	bmi	children	charges
count	1338.000	1338.000	1338.000	1338.000
mean	41.040	30.440	1.060	17663.520
std	13.530	5.980	1.220	11659.700
min	18.000	15.000	0.000	1100.000
25%	29.000	26.330	0.000	9834.220
50%	42.000	30.620	1.000	13979.670
75%	52.000	34.510	2.000	19331.440
max	64.000	47.260	5.000	57282.480

3. Active Production Model

Name : ridge (v1)
Algorithm : ridge
Trained : 2026-04-28 06:10:18
Dataset : medical_insurance.csv (1338 rows)

Metrics:
R2 = 0.9554
MAE = 1895.07
RMSE = 2434.31
CV R2 avg = 0.9505

4. Model Comparison (sorted by R2)

model	r2	mae	rmse	cv_r2_mean
ridge	0.955	1895.068	2434.309	0.951
lasso	0.955	1895.724	2437.503	0.951
linear	0.955	1895.768	2437.589	0.950
gbr	0.953	1987.850	2511.035	0.944
rf	0.949	2037.038	2604.398	0.941
xgb	0.938	2087.186	2876.524	0.938

5. Exploratory Data Analysis

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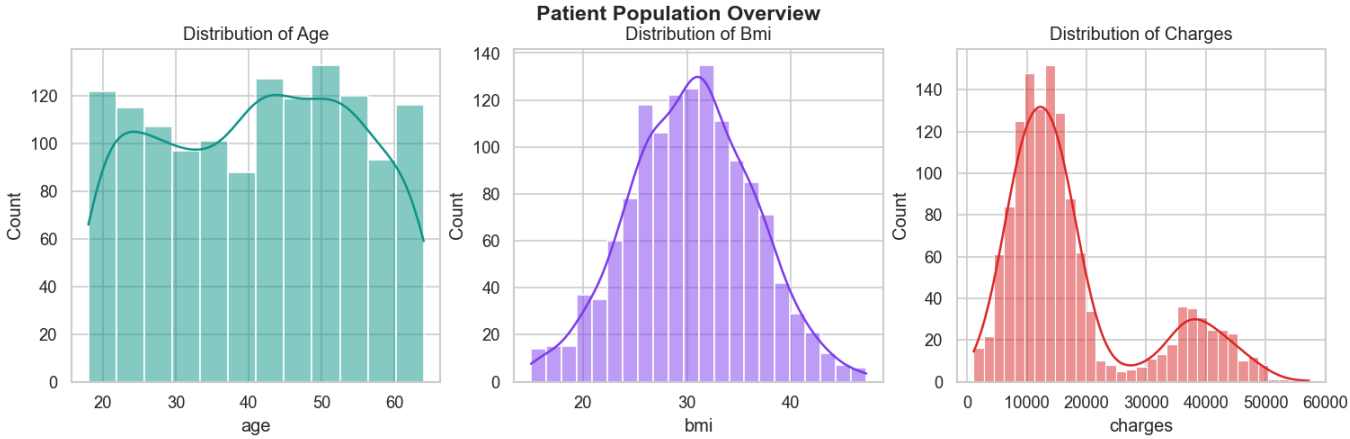
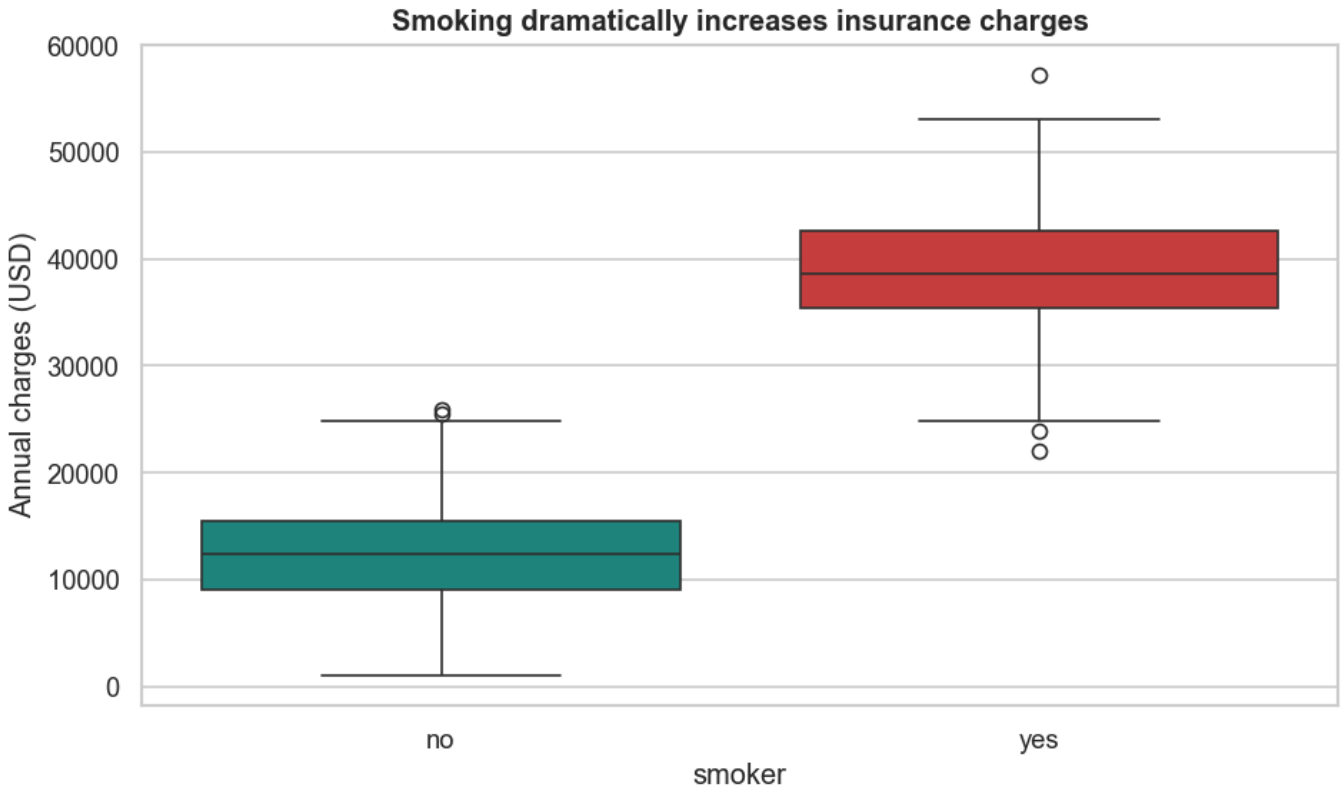
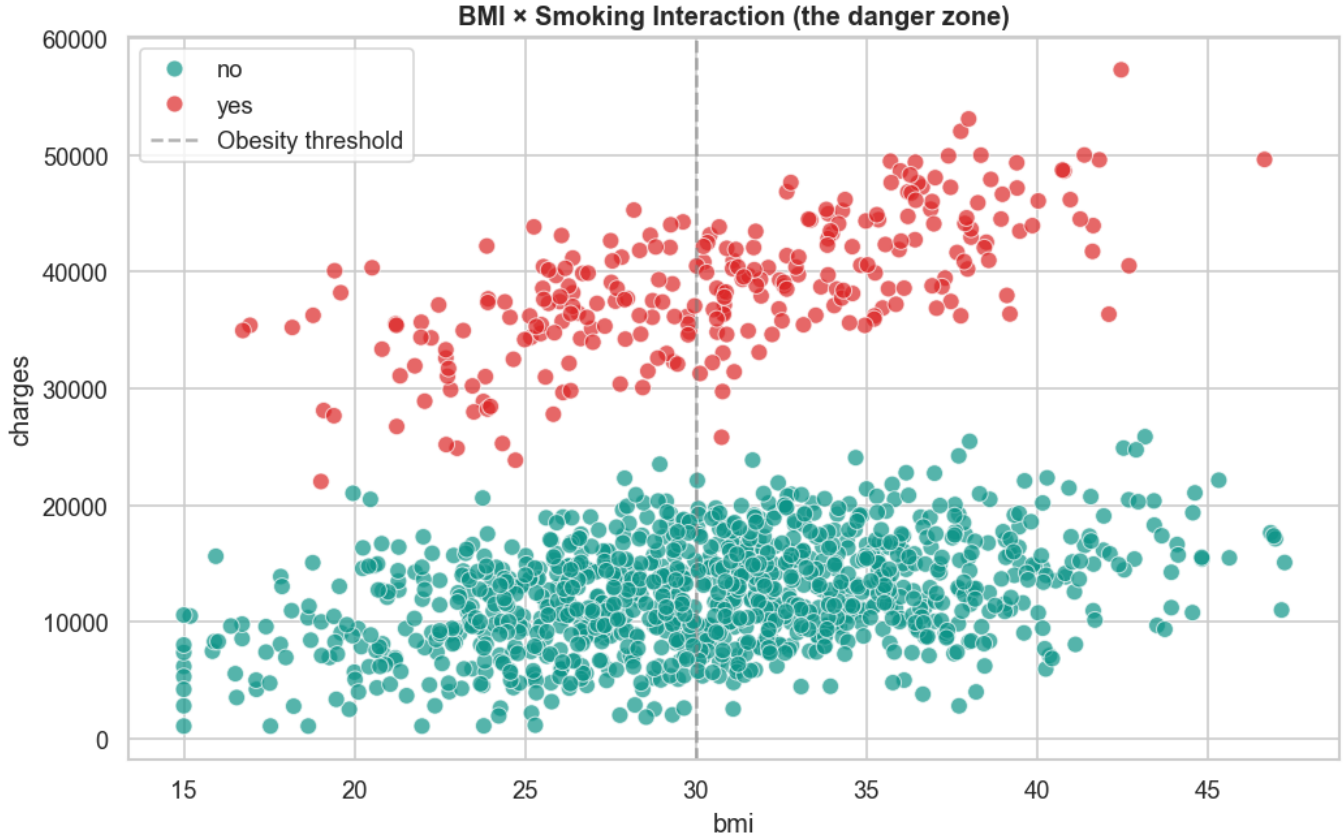
01 Distributions



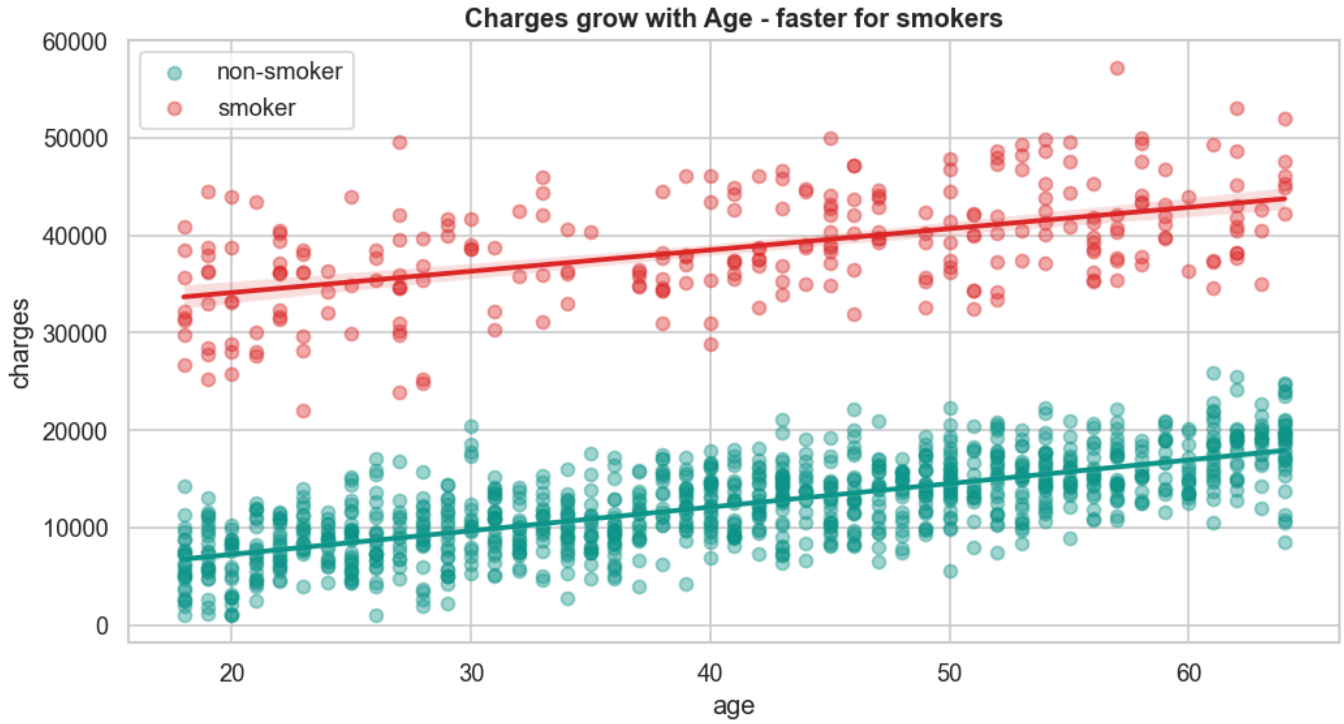
02 Smoker Impact



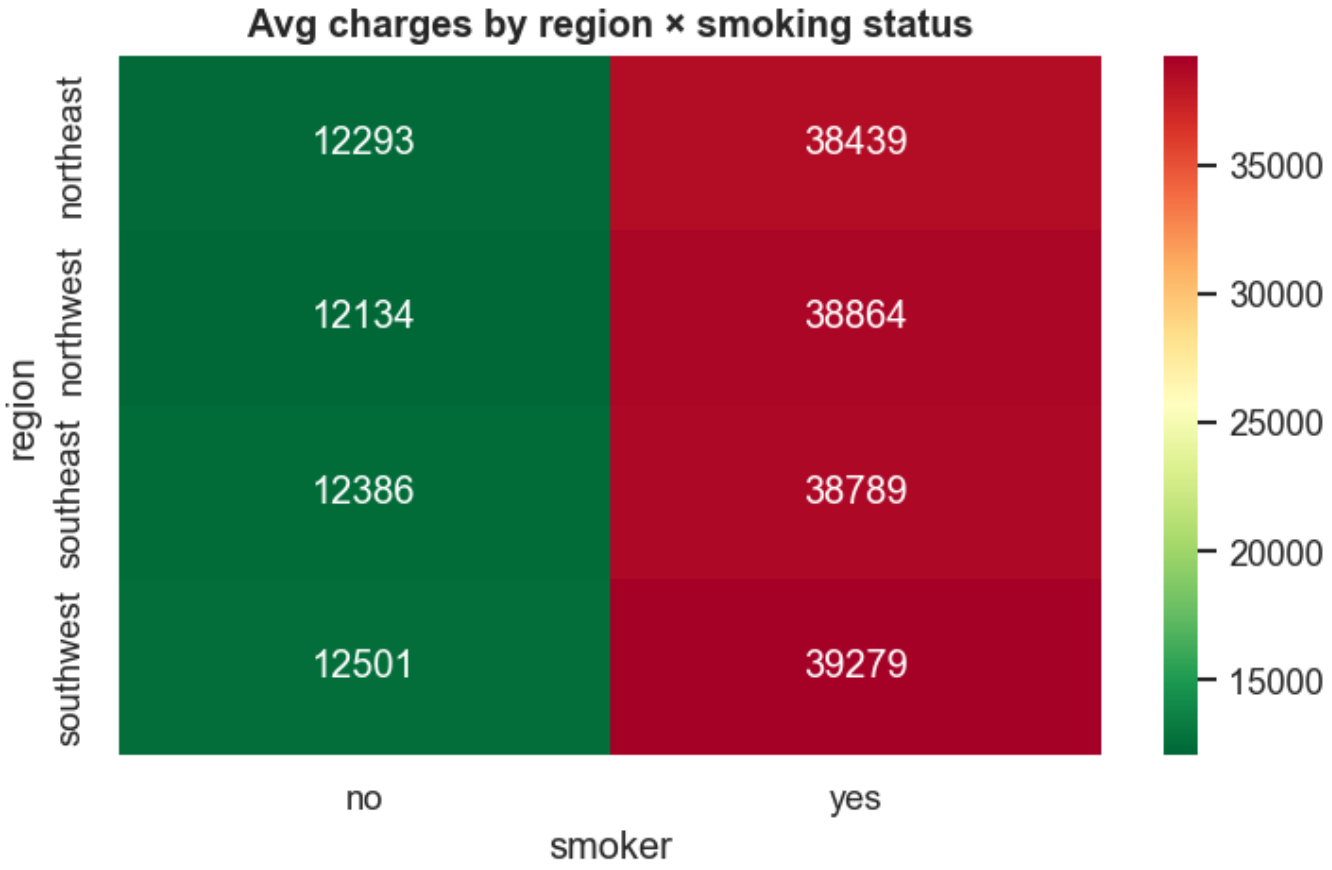
03 Bmi Smoker



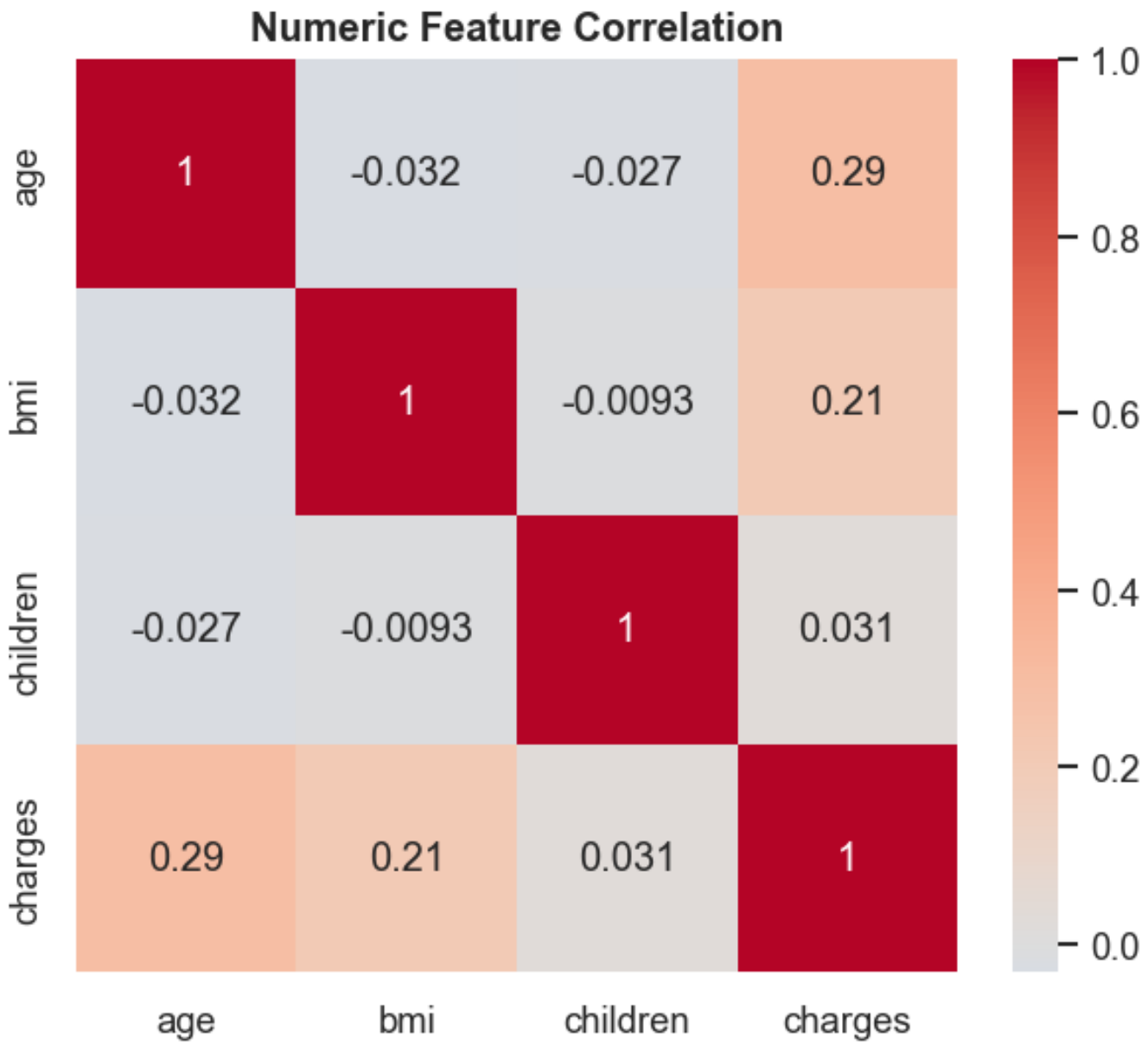
04 Age Trend



05 Region Heatmap



06 Correlation


Numeric Feature Correlation
age
bmi
children
charges
1
-0.032
-0.027
0.29
-0.032
1
-0.0093
0.21
-0.027
-0.0093
1
0.031
0.29
0.21
0.031
1
age
bmi
children
charges
1.0
0.8
0.6
0.4
0.2
0.0